

Iktos 3D Engine: Ligand and Structure-Based Drug Design at Scale

Iktos 3D Align, Iktos 3D Dock, and Boltz 3D Affinity

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Abstract

3D drug design in production spans two complementary regimes: **3D ligand-based (LB)** workflows that rank candidates on 3D similarities to a known active, and **3D structure-based (SB)** workflows that bring the protein pocket into the loop for pose prediction and affinity scoring. Teams routinely have a reference active, optionally a co-crystal pocket, and a decision to make : rank, refine, or promote.

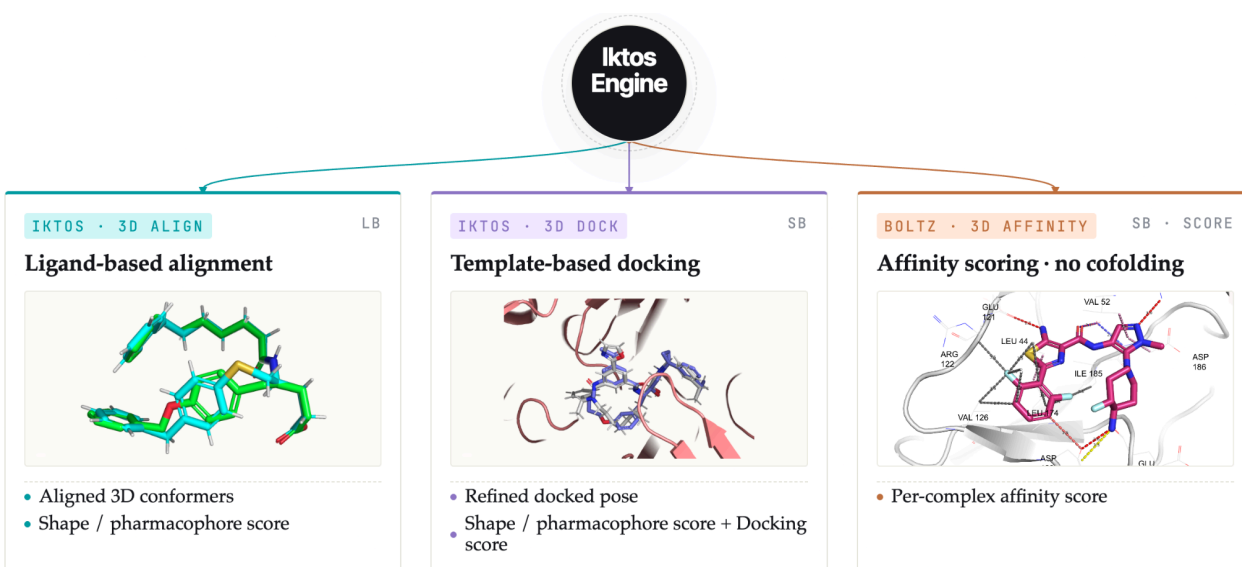


Figure 1 | Iktos 3D Engine: 3 focused APIs across Ligand Based & Structure Based regimes

We introduce the first version of *Iktos 3D Engine* ([Figure 1](#)) including three focused APIs that together cover both regimes at production scale:

- **Iktos 3D Align** : accurate ligand alignment method useful for 3D LB Virtual Screening.
- **Iktos 3D Dock** : accurate template-based docking method useful for 3D SB Virtual Screening.
- **Boltz 3D Affinity** : reuses the Boltz-2 affinity trunk ([Passaro et al. 2025](#)) directly on a user-supplied complex without (the time consuming) co-folding.

On their respective public benchmarks, the three APIs of *Iktos 3D Engine* compete with the published state of the art while running at production scale.

Introduction

3D drug design workflows share three operations that recur at every hit-to-lead optimisation project:

- **Pose alignment from a template** : align a given query ligand to a known reference ligand.
- **Pose prediction in the pocket** : produce a physically plausible pose well positioned in the pocket that downstream SB scoring functions can trust.
- **Affinity estimation on a trusted complex** : once a pose is in hand, estimate binding affinity.

We have integrated these three operations into *Iktos 3D Engine* as three composable APIs. Each API is scoped to a single operation and exposed as an independent service, letting project teams chain them into the workflow that matches their data. Each API is benchmarked on a public dataset suitable for its task, and each is engineered to meet the latency and budget constraints of real hit-to-lead optimisation projects.

Our key design choices are as follow :

- **Iktos 3D Align** : expert rather than neural, by design. Iktos team previously introduced the FMA-PO alignment framework in [Bergues et al. \(2025\)](#), achieving state of the art performance in molecular alignment and template-based docking. Iktos 3D Align is inspired by FMA-PO, retaining the Pose Optimisation (PO) part but replacing the neural pre-alignment step with a computational chemistry expert protocol based on common substructure matching. The goal is not an incremental gain in precision, but scalability and efficiency.

- **Iktos 3D Dock** : pocket-aware version of Iktos 3D Align. Iktos 3D Dock extends Iktos 3D Align by bringing the protein pocket into the loop: we augment the pose-optimisation (PO) objective of FMA-PO with a Vina scoring term and a soft-clash term, so that the optimised pose is not only consistent with the reference ligand but also physically plausible inside the pocket.
- **Boltz 3D Affinity** : Boltz-2 affinity trunk with no co-folding. We extract the Boltz-2 affinity head and serve it directly on user-supplied 3D complexes. This variant is not available in the upstream Boltz public repository. It removes the co-folding stage which is redundant whenever the user already trusts the input structure, leaving a pure affinity scorer that runs several times faster than the full co-folding pipeline.

Iktos 3D Align and **Iktos 3D Dock** are template-driven by design: they extract maximum signal from a known reference (an active molecule or a co-crystallised ligand) to constrain the 3D placement and refinement of every new candidate, which is the dominant operating regime in hit-to-lead optimisation.

The rest of this report presents each API in detail, benchmarks it against appropriate public datasets, quantifies its runtime, and provides examples of integrating them into a complete end-to-end workflow.

1. Iktos 3D Align

1.1. Method

3D ligand alignment is a classical problem with multiple existing tools (OpenEye ROCS ([Hawkins et al., 2007](#)) , ROSHAMBO ([Atwi et al., 2024](#)), LS-align ([Hu et al., 2018](#))...). **Iktos 3D Align** follows the FMA-PO framework, retaining the pose optimisation (PO) step while replacing the neural pre-alignment module (FMA) with an expert protocol combining graph-derived atom mappings and physics-based refinement.

The expert pre-alignment workflow generates multiple candidate correspondences through complementary substructure matching strategies ([Rascal MCES](#), flexible and strict [MCS](#), and hybrid combinations), ensuring robustness to chemical variability. Each mapping is used to produce an initial rigid alignment, followed by iterative force-field optimisation ([MMFF94](#)) under progressively refined constraints. The conformation with the highest 3D overlap score relative to the reference is retained.

Combining the expert pre-alignment workflow with PO yields a CPU-only protocol with a runtime of a few seconds per ligand. This design trades the incremental accuracy

that a neural pre-aligner may provide in edge cases for improved efficiency. In real-world hit-to-lead settings, where the template and query ligands are designed to share a common scaffold, this trade-off is favorable.

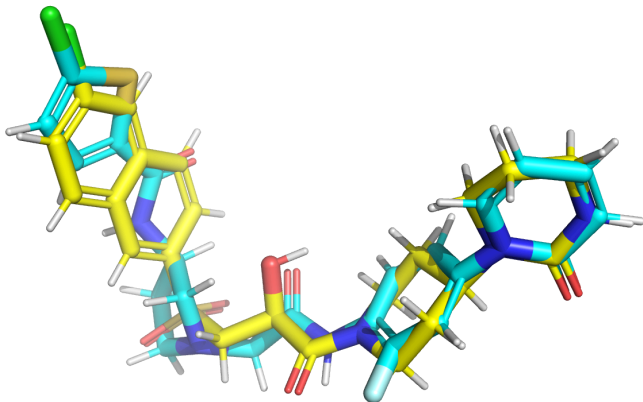


Figure 2: Iktos 3D Align pose prediction for PDB entry 2VVU (in cyan) using the ligand from PDB entry 3KL6 as template (in yellow)

1.2. Benchmark

We evaluate Iktos 3D Align on [AlignDockBench](#) ([Bergues et al. 2025](#)): 369 cross-docking pairs distributed over 61 anchor co-crystal complexes. For each pair, the reference and target pockets are aligned. Performance is reported as heavy-atom RMSD to the true co-crystal pose of the target ligand.

Threshold	< 1 Å	< 1.5 Å	< 2 Å	< 2.5 Å	< 3 Å
Iktos 3D Align	30.6 %	50.1 %	66.1 %	74.3 %	82.1 %
ROSHAMBO	10.0%	30.4%	48.5%	60.2%	68.3%
LSalign	14.9%	34.4%	54.2%	67.2%	75.6%

Table 1: Performance comparison of Iktos 3D Align and ligand-based methods on AlignDockBench.

Iktos 3D Align consistently outperforms ligand-based open source baselines across all RMSD thresholds (66.1% below 2 Å vs 48.5% and 54.2%), indicating higher accuracy in recovering near-native poses.

1.3. Use case : 3D ligand-based virtual screening

The primary use case for **Iktos 3D Align** is 3D ligand-based virtual screening ([Petrovic et al., 2022](#)): starting from a single reference active, we can rank candidates using the API by 3D shape similarity and 3D pharmacophore feature overlap with the reference. The API itself does not require a protein structure, only a reference ligand, which makes it the right tool whenever a target is unknown (cellular measured activity), its 3D structure unavailable or of low-quality. The ≈ 5 s per ligand CPU cost, combined with **Iktos 3D Engine** autoscaling, makes the approach practical for ligand-based virtual screening campaigns involving large libraries, as it supports the parallel processing of up to 15,000 compounds.

2. Iktos 3D Dock

2.1. Method

Unlike **Iktos 3D Align**, which does not use pocket information, ligand docking methods aim to place a ligand directly into a target binding site. These include classical docking approaches such as Autodock Vina ([Trott and Olson, 2010](#)), as well as hybrid template-based methods like FitDock ([Yang et al., 2022](#)) that leverage both a reference ligand and the protein pocket. In this context, **Iktos 3D Dock** is a template-based docking approach that uses the same expert pre-alignment as in **Iktos 3D Align** but extends the PO objective with two terms involving pocket information:

- A **Vina scoring term** ([Trott and Olson, 2010](#)) : a docking scoring function combining van der Waals interactions, repulsion, hydrophobic and hydrogen bonding terms.
- A **soft-clash term** : a smooth penalty on pairwise atom–atom overlaps with the protein.

2.2. Benchmark

On the same 369 AlignDockBench cross-docking pairs, **Iktos 3D Dock** shows performance very similar to **Iktos 3D Align** ([see Figure 4](#)), confirming that both approaches achieve comparable accuracy. They consistently outperform classical structure-based baselines such as FitDock, Autodock Vina and rDock, although they remain below FMA-PO, which achieve the best overall results.

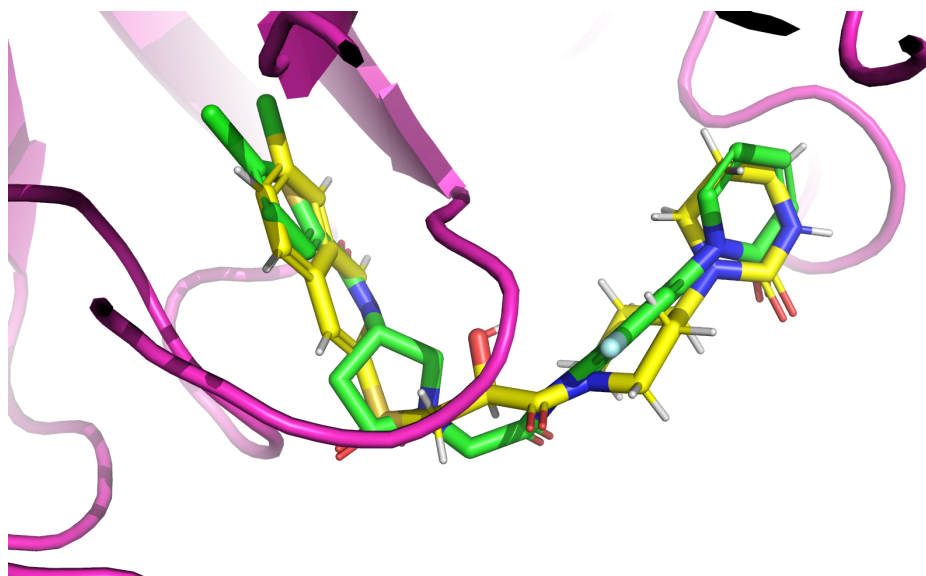


Figure 3 : Iktos 3D Dock pose prediction for PDB entry 2VVU (in green) using pocket and ligand from PDB entry 3KL6 as template (in magenta and yellow).

Importantly, Iktos 3D Dock and Align remain competitive while running on CPU, in contrast to FMA-PO methods that require GPU, highlighting a favorable trade-off between performance and computational cost.

Threshold	< 1 Å	< 1.5 Å	< 2 Å	< 2.5 Å	< 3 Å
Iktos 3D Align	30.6 %	50.1 %	66.1 %	74.3 %	82.1 %
Iktos 3D Dock	33.6%	50.4%	63.1%	75.1%	81.3%
FMA-PO	28.5%	53.9%	69.4%	77.8%	84.6%
FitDock	21.1%	36.0%	45.5%	50.4%	55.8%
Autodock Vina	20.3%	35.5%	47.2%	54.7%	59.1%
rDock	11.9%	24.7%	34.4%	42.0%	46.3%

Table 2: Performance comparison of Iktos 3D APIs and structure-based methods on AlignDockBench.

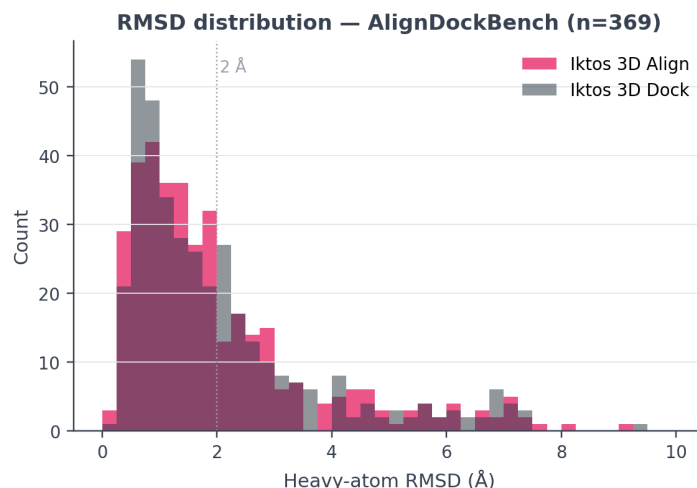


Figure 4 | RMSD distribution on AlignDockBench (369 cross-docking pairs).
Heavy-atom RMSD histograms for the IktoS 3D Align and the IktoS 3D Dock.

[Figure 5](#) shows the distribution of per-request latency across the 369 AlignDockBench cross-docking pairs, highlighting the computational cost of incorporating the protein pocket into the workflow. While **IktoS 3D Align** exhibits lower runtimes (median 4.7 s / ligand) than **IktoS 3D Dock** with a median approximately 6× larger (median 27.8 s / ligand). This overhead reflects the additional complexity of the docking stage, where pocket information is explicitly modeled, illustrating the trade-off between computational efficiency and physical realism ([see 2.3](#)).

A qualitative analysis is illustrated in [Figure 7](#): **IktoS 3D Align** and **IktoS 3D Dock** are compared to a panel of alignment and docking methods on three AlignDockBench cases, including a challenging macrocyclic-template case (PDB ID: 5QCY), where our approaches prove competitive.

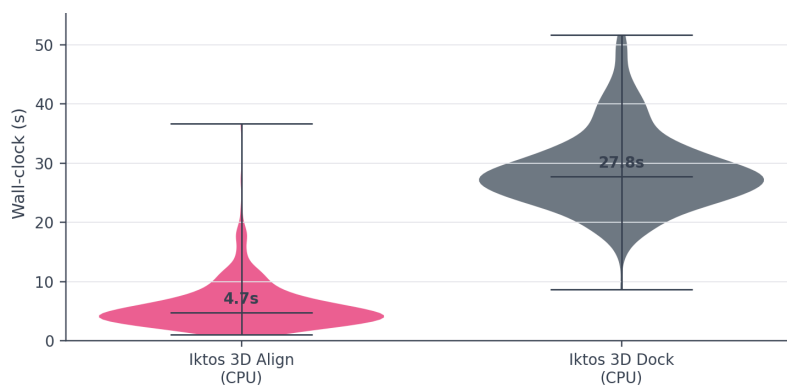


Figure 5 | Comparison of per-request latency on AlignDockBench. Wall-clock distribution over the 369 cross-docking pairs for IktoS 3D Align and IktoS 3D Dock on a single CPU core.

2.3. Physical plausibility : PoseBusters validity and ligand-pocket clash analysis

On AlignDockBench, **Iktos 3D Align** and **Iktos 3D Dock** achieve comparable RMSD performance, motivating additional experiments to better assess the impact of incorporating pocket information. We therefore evaluate each predicted pose against the [PoseBusters](#) test suite, the community-standard benchmark of structural-validity for docked poses, and report both the PoseBusters-valid (PB-valid) rate and the clash-specific criteria ([Figure 6](#)).

Iktos 3D Dock shows a substantial improvement in the overall PB-valid rate compared to **Iktos 3D Align**, increasing from 43.4% to 75.9% (+32.5 percentage points). This gain is primarily driven by pocket-aware constraints: the pass rate for minimum distance to the protein rises from 44.4% to 90.2% (+45.8 pts), while the volume overlap criterion reaches near-saturation at 99.7% (+13.6 pts).

At the same time, steric clashes with the protein are strongly reduced, with the median number of pairwise clashes dropping from 1 to 0 and the fraction of poses with at least one clash decreasing from 55.6% to 9.8% ([Figure 6](#), right). In contrast, intra-ligand quality metrics (bond lengths, ring flatness, aromatic planarity) remain consistently high ($\geq 98\%$) in both settings and contribute negligibly to the observed difference.

The gain in PB-validity justifies the additional runtime in a structure-based setting for **Iktos 3D Dock** compared to **Iktos 3D Align**, as it produces physically consistent, pocket-aware poses.

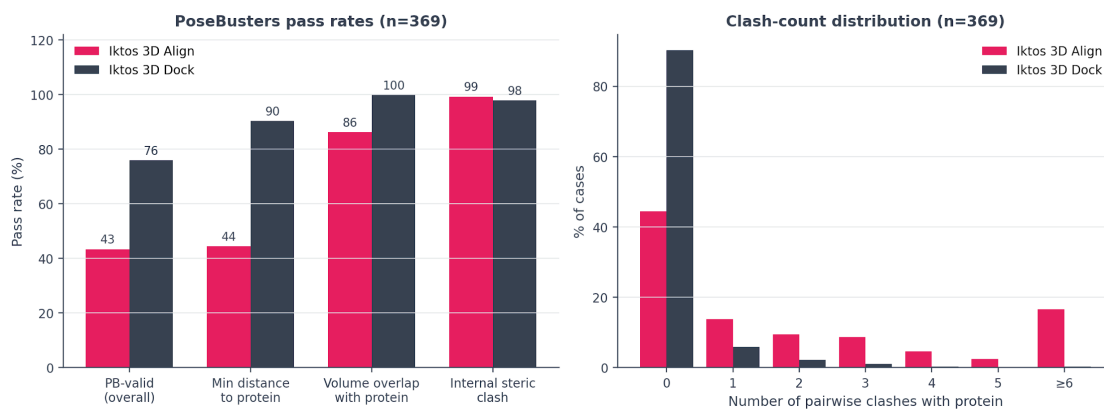


Figure 6 | PoseBusters validity on AlignDockBench (n=369). Left: pass rates on four PoseBusters criteria for Iktos 3D Align (magenta) and Iktos 3D Dock (grey). Right: distribution of the number of pairwise clashes with the protein.

2.4. Use case : structure-based virtual screening with template guidance

Where **Iktos 3D Align** serves ligand-based virtual screening, **Iktos 3D Dock** extends the same idea into structure-based virtual screening ([Anderson, 2003](#)). A query molecule is therefore selected on three criteria:

1. **3D shape similarity** to the reference molecule
2. **3D pharmacophore features overlap** with the reference molecule
3. **A favourable Vina docking score** in the targeted pocket

Effectively, **Iktos 3D Dock** isolates the subset of ligand-based hits that satisfy pocket constraints, delivering physically plausible poses ready for immediate downstream scoring.

The ≈ 28 s per ligand CPU cost, combined with **Iktos 3D Engine** autoscaling, makes the approach practical for structure-based virtual screening campaigns involving large libraries, as it supports the parallel processing of up to 10,000 compounds.

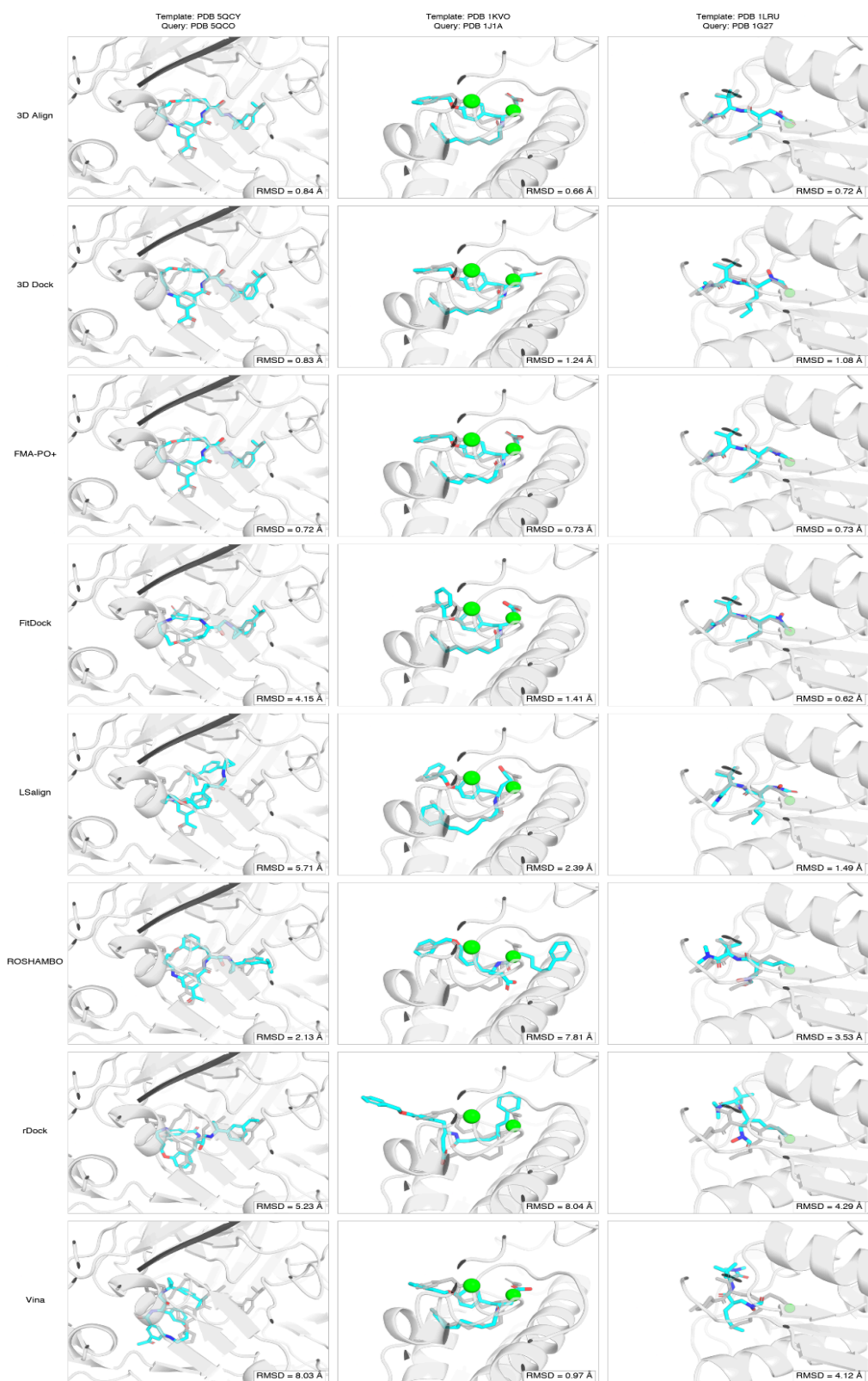


Figure 7 | Case studies on three AlignDockBench pairs. Predicted poses from Iktos 3D Align and Iktos 3D Dock and a panel of alignment and docking methods

3. Boltz 3D Affinity (co-folding-free)

3.1. Related work

Boltz-2 ([Passaro et al., 2025](#)) is a generative foundation model for protein–ligand structure and affinity prediction that, in its default co-folding mode, predicts the protein–ligand complex structure and estimates binding affinity using a dedicated affinity head. Boltz-2 represents the current state of the art in deep-learning-based small-molecule affinity ranking and significantly reduces the performance gap with gold-standard free-energy perturbation methods such as FEP+ ([Kuhn et al., 2017](#)) across multiple benchmarks while being 1000 times faster.

3.2. Method : co-folding is redundant when the pose is trusted

Production workflows typically start from a **trusted** 3D complex such as a co-crystal structure, a template-docked pose from [§2](#), or a molecular dynamics snapshot.

In this regime, the co-folding step of Boltz-2 is:

- redundant as it re-predicts a structure the user has already validated,
- a latency bottleneck as co-folding is the dominant cost of the Boltz-2 pipeline,
- a source of stochasticity that undermines reproducibility on ranking workflows,
- a potential source of errors (such as missing the right binding site or binding mode) which can undermine the user’s confidence in the affinity scoring.

Boltz 3D Affinity is a **co-folding-free** variant of Boltz-2: it directly reuses the Boltz-2 affinity head as published ([Passaro et al., 2025](#)), takes the user-provided protein structure in 3D and ligand pose as input, and omits the structure sampling modules.

The API returns two outputs from Boltz-2: the first corresponds to the native prediction from the binary affinity head, while the second is a transformation of the predicted continuous affinity into pIC50 value.

- `affinity_probability_binary` : the binder probability,
- `affinity_pIC50_pred_value = 6 – continuous_affinity_pred_value` : a convenience transform to the pIC50 scale.

This co-folding-free variant of Boltz-2 is not available in the upstream Boltz repository; it was implemented in-house and deployed in **Boltz 3D Affinity**.

3.3. Benchmark protocol

We evaluated **Boltz 3D Affinity** using the OpenFE subset of FEP+ Benchmark. Protein structures, docked ligands poses and experimental ΔG values were obtained from the Schrödinger public [FEP+ benchmark repository](#) (Ross et al., 2023). Systems were subsequently mapped to the canonical [OpenFE benchmark](#) system index in order to unify the naming conventions for structure and result files, and filtered to keep only OpenFE matching subsets. The final recovered dataset contains:

- **56 protein systems** organized into 8 subsets, each grouping systems by origin or chemical characteristics: `charge_annihilation`, `fragments`, `jacs`, `janssen`, `mcs_docking`, `merck`, `misc`, `water`.
- **1054 ligands with 3D poses**, of which 1026 have matching experimental ΔG values, corresponding to 97.3% coverage.

The reconstructed dataset contains more ligand structures than the OpenFE subset of FEP+ benchmark used in ([Passaro et al. 2025](#)), for the following reason: we retain all ligands present in the FEP+ structure files, regardless of their inclusion in any RBFE network, in order to maximize dataset coverage and provide a broader evaluation set.

Our analysis also highlights four targets (CDK2, JNK1, P38, and TYK2), in line with [Passaro et al. \(2025\)](#), due to their extensive use in affinity prediction benchmarks.

The evaluation metric is the per-protein-system Pearson correlation between predicted and experimental pIC50 values, averaged across systems and weighted by the number of ligands, following the protocol of [Passaro et al. \(2025\)](#). Each ligand is evaluated five times independently, and final predictions are obtained by averaging the results across runs.

We compared two modes of Boltz-2 affinity prediction:

- **Co-folding-free mode** : **Boltz 3D Affinity**, scores the user-supplied 3d structures of ligand and protein.
- **Co-folding mode** : the standard Boltz-2 pipeline from the upstream Boltz repository, running on the same infrastructure for fair comparison.

3.4. Primary results

[Figure 8](#) and [Table 3](#) show the weighted Pearson correlation computed on the full FEP+/OpenFE Benchmark and on the 4 selected targets (CDK2, JNK1, P38, and TYK2).

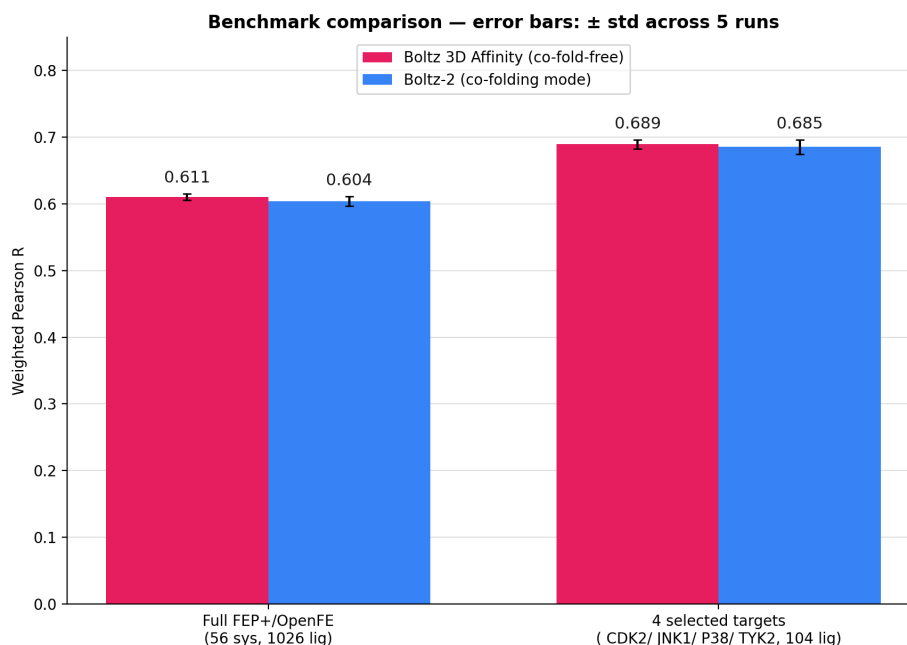


Figure 8 | Co-folding-free Boltz 3D Affinity vs Boltz-2 co-folding mode. Benchmark summary showing the weighted Pearson correlation for the co-folding-free Boltz 3D Affinity and Boltz-2 co-folding mode on the full FEP+/OpenFE benchmark and on the 4 selected targets.

Subset	Boltz 3D Affinity (co-fold-free)	Boltz-2 (co-folding mode)
Full Benchmark (56 systems, 1026 ligands)	0.611 $\pm$ 0.004	0.604 $\pm$ 0.007
4-target subset (jacs) (CDK2 / JNK1 / P38 / TYK2, 104 ligands)	0.689 $\pm$ 0.006	0.685 $\pm$ 0.010

Table 3 | Pearson correlation comparison of Boltz-2 and Boltz 3D Affinity on FEP+/OpenFE benchmark.

[Figure 9](#) shows per-target Pearson correlation on the 4-target subset:

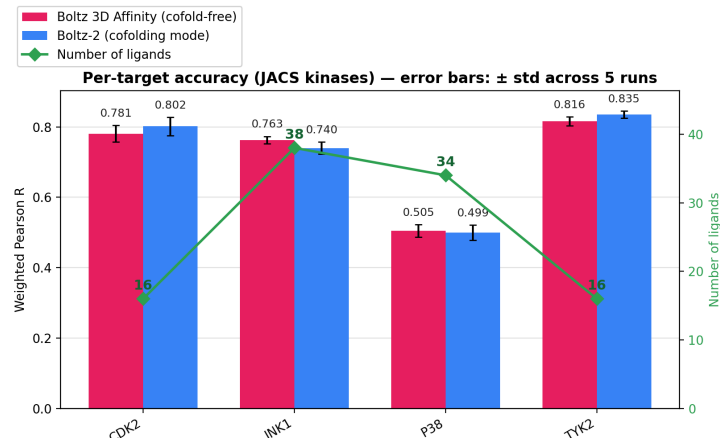


Figure 9 | Per-target affinity Pearson correlation on CDK2, JNK1, P38, and TYK2 of jacs subset.

[Figure 10](#) shows the weighted Pearson correlation per OpenFE subset including the number of systems and ligands per subset.

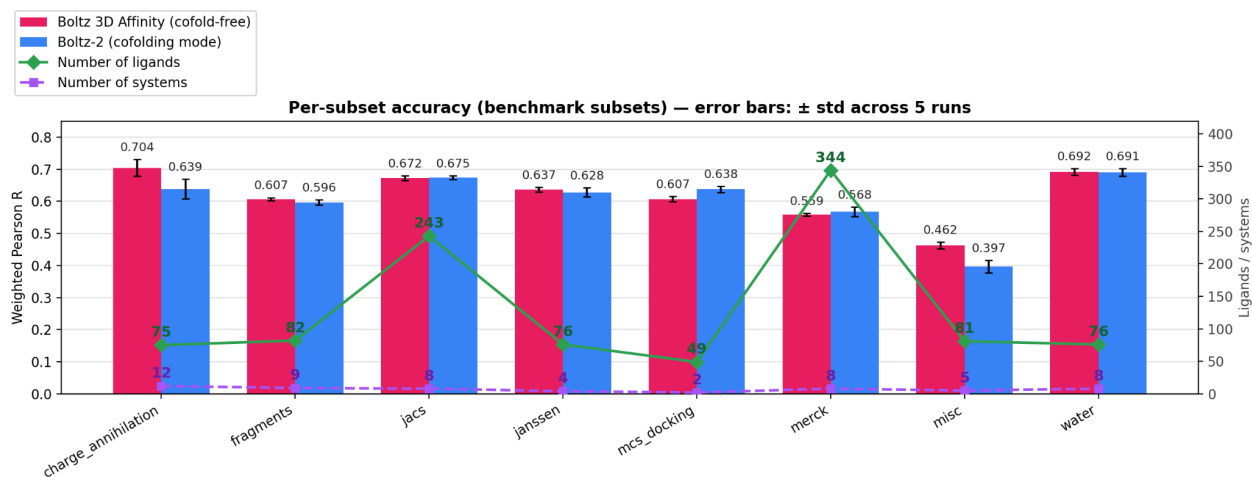


Figure 10 | Weighted Pearson correlation across OpenFE subsets with number of systems and ligand count for each subset.

3.5. Co-folding-free $\approx$ co-folding : the core operational claim

[Figure 8](#) and [Table 3](#) shows that the co-folding-free Boltz 3D Affinity is statistically indistinguishable from Boltz-2 in its default co-folding mode on both sets; the difference between the two means is well within the per-run standard deviation.

The same conclusion is supported by [Figure 9](#) and [Figure 10](#). Analyses by subset and by target both indicate that the two models are globally equivalent in performance.

Across all 1,026 ligands, the co-folding-free **Boltz 3D Affinity** and Boltz-2 (default co-folding mode) predictions are highly correlated (Pearson $r = 0.991$) and differ on average by only 0.11 pIC50 units ([Figure 11](#)).

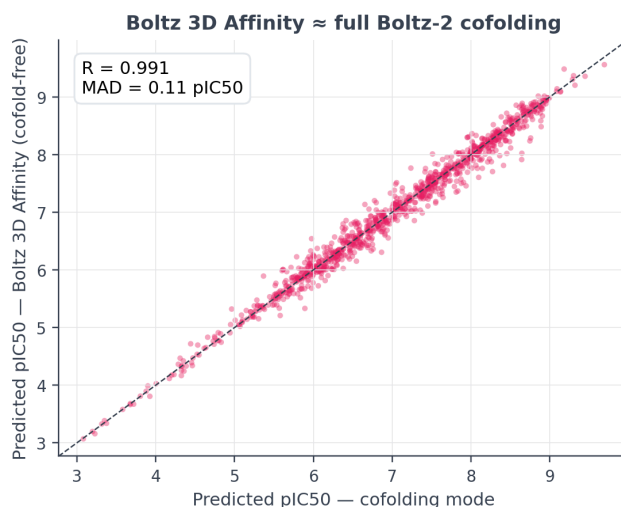


Figure 11 | Co-folding-free Boltz 3D Affinity vs Boltz-2 co-folding mode.

In other words, once a reliable 3D complex is available (e.g. from an accurate template based docking method like **Iktos 3D Dock**), co-folding provides negligible improvement to affinity prediction while still imposing a substantial computational cost, which is exactly the kind of overhead production workflows seek to eliminate.

3.6. Runtime

Runtime was benchmarked across the 1,054 ligands of the FEP+/OpenFE benchmark using both the co-folding-free **Boltz 3D Affinity** pipeline and the Boltz-2 co-folding approach.

Wall-clock time scales with the size of the protein-ligand complex, quantified by a token count (sum of protein residues and ligand heavy atoms; [Figure 12](#), right). Small complexes (< 300 tokens) run in a median of ~14 s of GPU time, while medium-to-large complexes (300–700 tokens) saturate around 19–20 s of GPU time per complex.

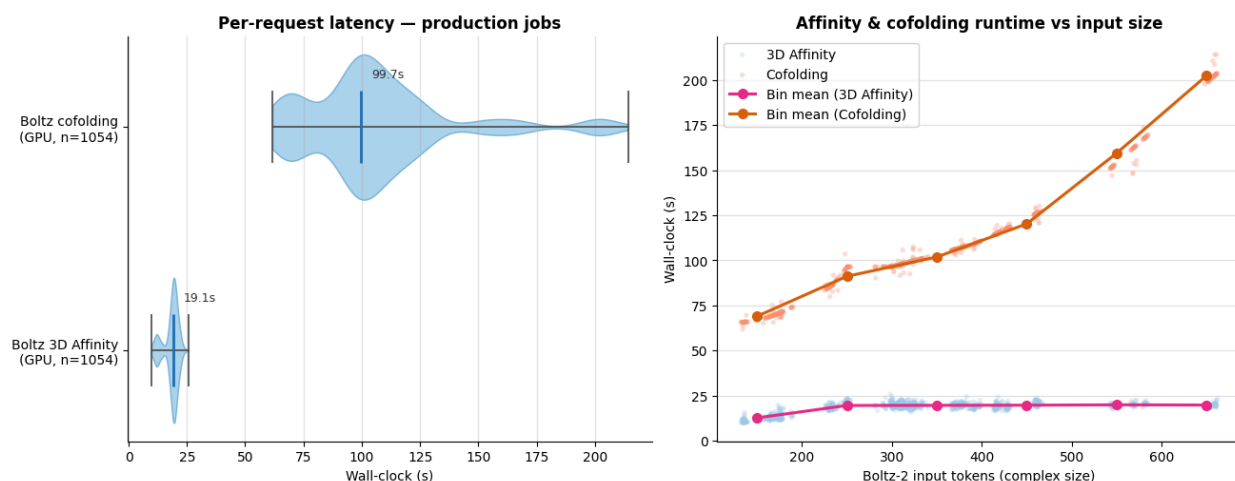


Figure 12 | Boltz 3D Affinity and Boltz co-folding runtime on production traffic. Left: wall-clock distribution on 1054 production jobs. Right: wall-clock as a function of complex size measured by Boltz-2 input tokens.

By comparison, the full co-folding mode on the same complexes typically takes several minutes of GPU time per ligand on the same hardware with runtime continuing to increase as the size of the protein-ligand complex grows. The co-folding free **Boltz 3D Affinity** therefore delivers a significant speed-up at no measurable accuracy cost on the benchmark.

4. Composition strategies for virtual screening

The three APIs can be used as a chained workflow, enabling a project team to start from a known active, and a 3D protein, and execute a complete virtual screening pipeline end to end.

Two cascade funnel strategies can be used ([Figure 13](#)) and are detailed below.

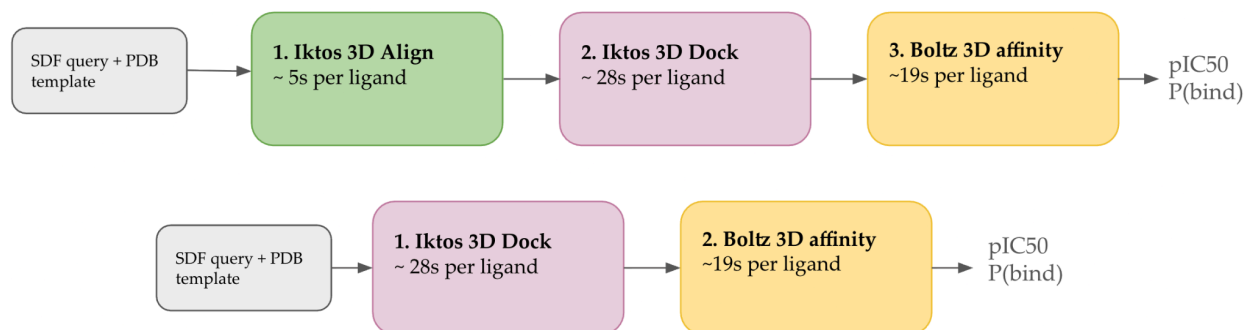


Figure 13 | End-to-end Iktos 3D API workflows. A user-supplied PDB template and an SDF query are routed through Iktos 3D Align, Iktos 3D Dock, and Boltz 3D Affinity.

4.1. Cascade A : Iktos 3D Align → Iktos 3D Dock → Boltz 3DAffinity

The full three-stage funnel is the appropriate default when working with large input libraries (10^5 – 10^6 compounds), where computationally inexpensive early filtering is advantageous and a final affinity estimate is required :

1. **Iktos 3D Align**: 3D ligand-based virtual screening first pass. Each query molecule is aligned to the reference active ligand and scored on shape and pharmacophore overlap. Cheap (≈ 5 s / ligand on CPU). Top-k by 3D ligand-based score is forwarded to stage 2.
2. **Iktos 3D Dock**: pocket-aware refinement. The shortlisted poses are refined under pocket constraints and re-ranked on a composite of shape, pharmacophore and Vina docking score. Top-k by 3D structure based score is forwarded to stage 3.
3. **Boltz 3D Affinity**: Boltz-2 affinity scoring is applied to the refined complexes for the final ranking providing activity probabilities and pIC50 estimates.

4.2. Cascade B : Direct: Iktos 3D Dock → Boltz 3D Affinity

We may skip **Iktos 3D Align** for a pure 3D structure based pipeline and start directly at the **Iktos 3D Dock** stage. This is the right default whenever **every input compound is worth a pocket-aware pose**.

5. Reproducibility

All the experiments reported in this manuscript can be reproduced using the available code and data in the following github repository : <https://github.com/iktos/engine>

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